The downloaded data is in single CSV files. We use Pandas to read the datasets and Spark to write to a Delta table. We demonstrate this for only the first file in the following code block (*Figure 3.17*).

```
df = pd.read_csv(f'{cloud_storage_path}/holidays_events.csv', sep=',', decimal='.')
df['date'] = pd.to_datetime(df['date'])
spark.createDataFrame(df).write.mode("overwrite").saveAsTable("holiday_events")
display(df)
```

Figure 3.17 – Reading in the sales holiday events datasets with Pandas

Each table for the Favorita project is transformed into Delta tables similarly. As a result, we only include the first table in the book's pages. However, the code that transforms each table is in the repository.

## Project: Retrieval Augmented Generation Chatbot

**RAG** stands for **Retrieval Augmented Generation**. A RAG system often consists of your data, a vector database, a search algorithm, and a **generative AI** model to generate an answer to a user's query. *Figure 3.18* shows the pipeline we will build throughout this book.

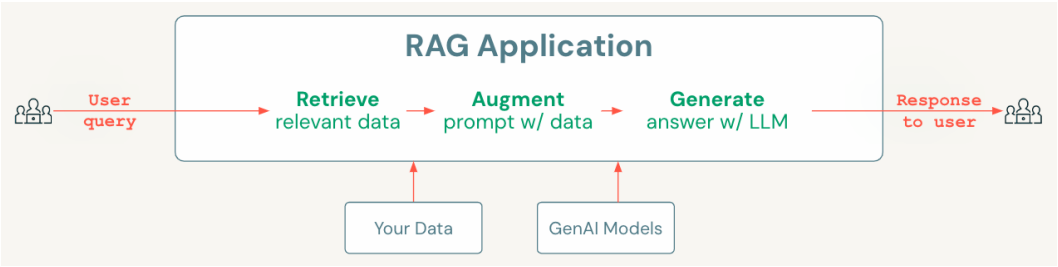



Figure 3.18 – A RAG example pipeline flow diagram

We see from *Figure 3.18* that we require retrieving relevant data according to the user query. That is where our search method would access a vector database and conduct a semantic or hybrid search. At this stage in the RAG Chatbot project, we have our PDFs in their raw, unstructured form. We want our users to access this knowledge via our chatbot, so we must bring relevant content from the PDFs into it, running in real-time. To do so, we'll convert all our PDFs into a machine-readable format. This chapter shows how to extract unstructured text from PDF files, chunk your data, and transform your text into embeddings. Then, we store the embeddings in a Delta table.